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April 8, 1964
SUPERSEDING
Fed. Spec. GGG-B-485
April 11, 1962

FEDERAL SPECIFICATION

BLOCK, TACKLE, WIRE ROPE, WITH SWIVEL HOOK; (FOR CRANE USE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single and multiple sheave wire rope hook blocks with swivel hook for use with locomotive, crawler-, truck-, and wheel-mounted cranes.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Sizes.* Blocks shall be of the following sizes and capacities, as specified (see 6.1):

Size		*Capacity*
5.	—	5-ton
10.	—	10-ton
20.	—	20-ton
25.	—	25-ton
35.	—	35-ton
40.	—	40-ton
55.	—	55-ton
70.	—	70-ton
75.	—	75-ton

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-S-890—Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL-F-3541—Fittings, Lubrication.

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MIL-P-3865—Packaging and Packing for Overseas Shipment of Blocks (Wire and Manila Rope).

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-271 — Nondestructive Testing Requirements for Metals.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Governmental:

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Copies may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Description. Each block shall consist of one or more steel sheaves, steel side plates, separator plates, swivel hook, bearings, becket, and all other parts necessary for satisfactory operation. Adequate keepers shall be provided to prevent rope from leaving sheaves and being damaged during operation. The block shall be new and unused.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of good commercial quality entirely suitable for the purpose intended. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.3 Interchangeability. Replaceable parts shall be manufactured to standard tolerances to permit replacement without requiring modification.

3.4 Design. The block shall conform to table I. The design shall be based on a factor of safety of not less than 5. The block shall be capable of withstanding the test loads without breakage or permanent distortion of any component part.

3.5 Sheaves. Sheaves shall be of high grade medium carbon cast steel, manganese cast steel, or rolled or welded structural steel. Sheaves shall be designed for use with the wire rope size specified (see 6.1). The outside diameter of the sheave shall be not less than 18 times the diameter of the wire rope for which the sheave is designed. Sheave grooves shall be designed to give the rope a 150° arc of support with

TABLE I.—*Design characteristics*

Capacity (safe working load in 2000-pound tons)	Proof test load (2000-pound tons)	Number of sheaves	Overall length not more than (inches)	Gross weight not less than (pounds)	Hook opening not less than (inches)
5	10	1	31	100	2
10	20	2	36	200	2
20	40	2	42	430	3
25	50	3	42	500	3
35	70	3	48	575	3-1/4
40	80	3	56	900	3-1/2
55	110	3	63	1,000	4
70	140	3	64	1,250	4
75	150	4	65	1,300	4

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the sides of the groove tangent to the ends of the bottom arc. The depth of the groove shall be not less than 1.5 rope diameters. Sheave grooves shall be machined true with the sheave bore. When medium carbon cast steel or structural type steel plate is furnished for sheaves 12 inches in diameter and larger, the sheave groove shall be hardened to a minimum of Rockwell C-35. Sheave hubs shall be machined and shall be designed to bear against the side plates and separator plates except where bearings take the end thrust.

3.6 Sheave pin. Sheave pins shall be precision type alloy or carbon steel shafting, turned, ground, and polished except that when an inner bearing race is used, the sheave pin need not be polished. The pin shall be designed to transmit the load to the frame. Suitable means shall be provided to prevent rotation of the pin.

3.7 Swivel hook. The swivel hook shall be forged steel. For Navy use, the steel shall conform to any class in MIL-S-890 with a longitudinal elongation in 2 inches of not less than 20 percent. Unless otherwise specified (see 6.1), the hook shall be the safety type fitted with a safety latch actuated by a stainless steel or other corrosion resistant spring. The hook shall be suspended from the block by means of a yoke or cross-head and shall swivel freely through 360° with not less than the applicable safe working load specified in table I suspended from the hook. There shall be no cracks or other discontinuities in the finished hook. The narrowest part of the hook opening shall be not less than that shown in table I. When specified (see 6.1), the material designation and heat treatment shall be marked on the hook using combinations of the following applicable symbols and appropriate numbers: SAE (Society of Automotive Engineers), AISI (American Iron and Steel Institute), TS (tensile strength), H (heat with degrees given), N (normalized), A (anneal), OQ (oil quench), WQ (water quench), AQ (air quench), RH (reheat with degrees given), FC (furnace cool), and AC (air cool).

3.8 Bearings.

3.8.1 *Sheave bearings.* Each sheave shall be mounted on ball bearings or two rows of short roller bearings or a full roller bearing. Bearings shall have suitable retainers or seals for retention of lubricant and exclusion of foreign matter.

3.8.2 *Swivel hook bearings.* Hooks shall swivel on one or more ball or roller bearings. Bearings shall be of such size and quantity as to permit swiveling of the applicable safe working load specified in table I without twisting or opening up the wire rope. The bearings shall be adequately sealed for retention of lubricant and exclusion of foreign matter.

3.9 Becket. Each block shall be supplied with a suitable becket. The becket shall show no evidence of breakage or permanent distortion with the rope dead ended at the becket and the applicable proof load specified in table I suspended from the hook.

3.10 Cheek-plate weights. Cheek-plate weights, when utilized, shall be of steel or cast iron for blocks having sheaves of 12 inches or less diameter. Plates shall be designed with recesses for boltheads and nuts.

3.11 Lubrication. Sheave and swivel hook bearings shall be provided with suitable means for lubrication. Hydraulic lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Each sheave bearing shall have an independent lubricating line located within the sheave pin.

3.12 Treatment and painting. All metal surfaces except bearing surfaces and hooks shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice. The color shall be in accordance with Fed. Std. No. 595, color No. 13538, yellow gloss.

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3.13 Identification markings. Each block shall be marked with the following information:

(*a*) Manufacturer's name or trademark.
(*b*) Manufacturer's part number.
(*c*) Safe working load in tons.
(*d*) Block size (i.e., sheave diameter).

The marking shall be applied to an identification plate (nameplate) which is securely fastened to the block, or may be applied directly to the surface of the block by acid or electric etching, engraving, embossing, casting, or molding. Marking shall be on the side of the block and shall be of such size and type as to be clearly legible after painting.

3.14 Workmanship.

3.14.1 *Steel fabrication.* Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shearing or sawing. Reentrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.14.2 *Bolted connections.* Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight. All screw threads shall comply with National Bureau of Standards Handbook H28. Unified screw threads shall be used in lieu of American National screw threads and non-unified American Standard screw threads.

3.14.3 *Riveted connections.* Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and shall be in full contact with the surface of the member.

3.14.4 *Welding.* The surfaces of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Before assigning any welder to manual welding work, the manufacturer shall certify to the contracting officer that the welder has passed the qualification tests as prescribed by either the American Welding Society or the American Society of Mechanical Engineers for the type of welding operation to be performed.

3.14.5 *Machine work.* Tolerances and gages for metal fits shall conform to the limitations specified herein and otherwise to the standards of good commercial practice.

3.14.6 *Castings.* Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the castings unsound for use.

3.14.7 *Forgings.* All forgings shall be of uniform quality, free of detrimental flash, scale, cracks, hard spots, cold shuts, and excessive cooling stresses. Strength and other essential physical properties of the forgings shall be adequate throughout for the purpose intended.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as other-

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wise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Examination.** Each block shall be examined to verify compliance with this specification. Corrections required, following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. Examinations shall be conducted as specified in table II. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE II.—*Classification of defects in accordance with MIL-STD-105.*

Examine	Requirement paragraphs	Defects
Material	3.2	Material used in fabrication or manufacture of any component or part not as specified, obviously inferior, or not suited to the purpose.
Design	3.4	Any applicable dimension or capacity not as specified in table I.
Sheaves	3.5	One or more dimensional characteristics not as specified. Hardness of groove does not meet minimum requirements.
Sheave pin	3.6	Sheave pin not as specified. Pin rotates.
Swivel hook	3.7	Any unauthorized deviation from or circumvention of one or more specified details.
Bearings	3.8 through 3.8.2	Not in accordance with specified requirements.
Becket	3.9	Missing or inadequate.
Cheek-plate weights	3.10	Not recessed for boltheads and nuts.
Lubrication	3.11	Means for lubrication not provided in one or more locations where lubrication is required or essential. One or more fittings missing, damaged, or not according to referenced specification. Provisions for sheave bearing lubrication not as specified.
Treatment and painting	3.12	Color of finish not as specified. Finish blistered, peeled, chipped, or otherwise impaired.
Identification marking	3.13	Missing or not affixed in prescribed manner, or insecurely fastened. Data incomplete, illegible, or erroneous.
Bolted and riveted connections	3.14.2 and 3.14.3	One or more missing, loose, deformed, or of a size not suited to the purpose.
Welding	3.14.4	Poor appearance, incomplete, porous, sparse, flux inclusions, excessive spatter, undercuts or lack of fusion with parent metal or between parts.
Machine work	3.14.5	Surfaces rough, pitted, uneven, or excessively tool marked. Any machined surface condition adversely affecting tolerances or fitups of mating or working parts.
Castings	3.14.6	Blowholes, patches, warps, extraneous fins, or any other similar or characteristic defect impairing the function, operability, reliability, or appearance of the piece, part, component, or assembly.
Forgings	3.14.7	Presence of cracks, laps, cold shuts, or any other malformation or defect impairing the intended use or serviceability.

4.3 **Sampling procedures for tests.**

4.3.1 *Lot size.* All complete production units of the same size, produced under the

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same manufacturing processes and conditions, at the same plant, and which are scheduled for delivery to the Government at one time, shall be divided into lots of not more than 100 units.

4.3.2 *Samples.* Random samples of production units shall be selected from each lot for testing. The selection, handling, and disposition of samples shall be according to the provisions of MIL-STD-105 at inspection level S-4. Samples shall be subjected to the overload test specified in 4.4.3 and, when specified (see 6.1), to the service test specified in 4.4.4. The Acceptable Quality Level shall be 2.5 percent defective.

4.4 **Tests.** All blocks produced in fulfillment of a Government contract or order shall be tested as specified in 4.4.1 and 4.4.2. Test specified in 4.4.3 and 4.4.4 shall be conducted under the sampling procedures specified in 4.3. Failure to pass any phase of the tests shall constitute cause for rejection.

4.4.1 *Proof test.* Each complete block shall be reeved, without using becket, with wire rope of the size recommended for use thereon. The applicable proof test load specified in table I shall be suspended from the hook for a period of not less than 10 minutes. The block shall then be reeved with the wire rope dead ended at the becket and the load reapplied for 10 minutes. The becket shall be examined to determine conformance to 3.9.

4.4.2 *Hooks.* Each hook shall be subjected to a magnetic particle test in accordance with MIL-STD-271 and shall conform to 3.7 prior to assembly.

4.4.3 *Overload test.* The block shall be reeved, without using the becket, and a load equal to 1-1/2 times the applicable safe working load specified in table I suspended from the hook. The load shall then be raised and lowered a minimum of 20 feet at speeds selected by the manufacturer.

4.4.4 *Service test.* When specified (see 6.1), the sample block shall be subjected to the following service test. With the block reeved as in 4.4.3, a load equal to the applicable safe working load specified in table I shall be raised and lowered a minimum of 20 feet. The cycle shall be repeated a minimum of 10 times at speeds approximating those encountered in crane service. The line speed shall be not less than 120 feet per minute. The load shall be rotated through 360° a minimum of three times to determine conformance to 3.8.2.

4.5 **Preparation for delivery inspection.** The preservation, packaging, packing, and marking of the blocks shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of preservation, packaging, and packing.)

5.1 **Preservation and packaging.** Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.* The blocks shall be preserved and packaged in accordance with MIL-P-3865.

5.1.2 *Level C.* The blocks shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Levels A and B.* The blocks shall be packed in accordance with MIL-P-3865.

5.2.2 *Level C.* Blocks shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

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5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order (see 6.1), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order (see 6.1), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size of block required (see 1.2.1).
(*c*) Wire rope size for sheaves (see 3.5).
(*d*) When nonsafety type hook is required (see 3.7).
(*e*) When material designation and heat treatment markings are required on the hook (see 3.7).
(*f*) When service test is required (see 4.3.2 and 4.4.4).
(*g*) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(*h*) Special marking when required (see 5.3.1 and 5.3.2).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tackle blocks, snatch blocks, or sheaves, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Tackle blocks, snatch blocks, or sheaves, not otherwise indexed.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MO (ASMC)

Navy—Docks

Air Force—WRA

Interested Military Activities:

Review

Army

Navy—YD

Air Force—84

User

MO

WP, CG, MC

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